# Class 5: Data Visualization with ggplot2

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## Overview

In this lab we explore data visualization using the **ggplot2** package in R. Every ggplot requires at minimum: a dataset (`ggplot()`), aesthetic mappings (`aes()`), and a geometric layer (e.g. `geom_point()`).

## Loading ggplot2

We load ggplot2 before use. Note: `install.packages()` is run once in the console only — never left uncommented in a document.

```
# install.packages("ggplot2")
library(ggplot2)
```

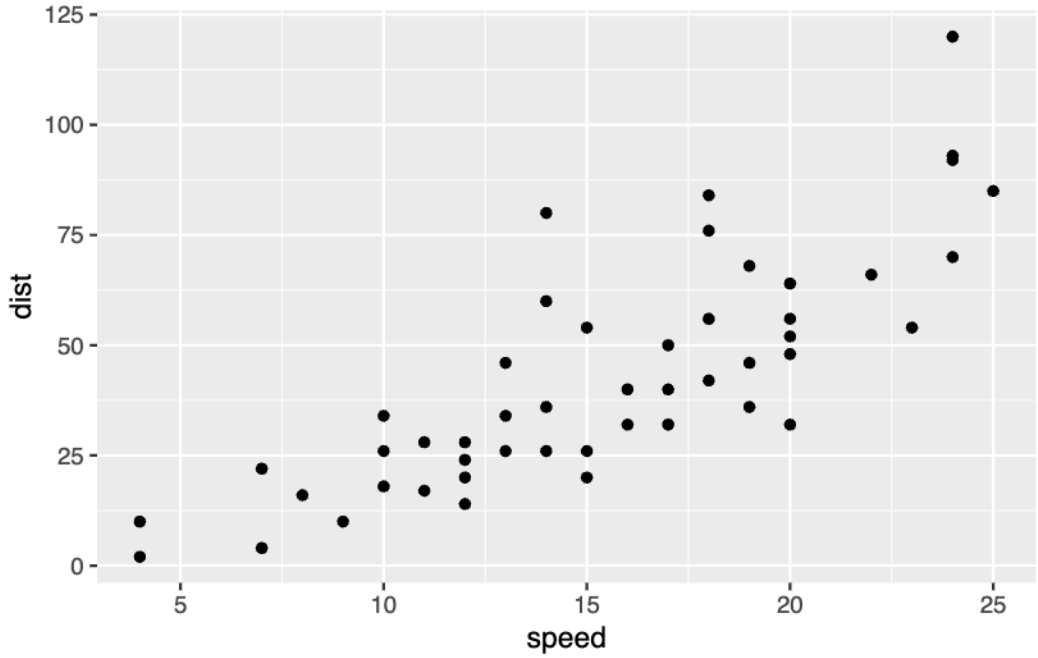
---

## Section 6: Creating Scatter Plots

### First geom_point() plot — `cars` dataset

The built-in `cars` dataset has two variables: `speed` and `dist` (stopping distance).

```
# Basic scatter plot of cars dataset using ggplot + aes + geom_point
ggplot(cars) +
  aes(x = speed, y = dist) +
  geom_point()
```




Each point represents a car. Faster cars take longer to stop — a clear positive relationship.

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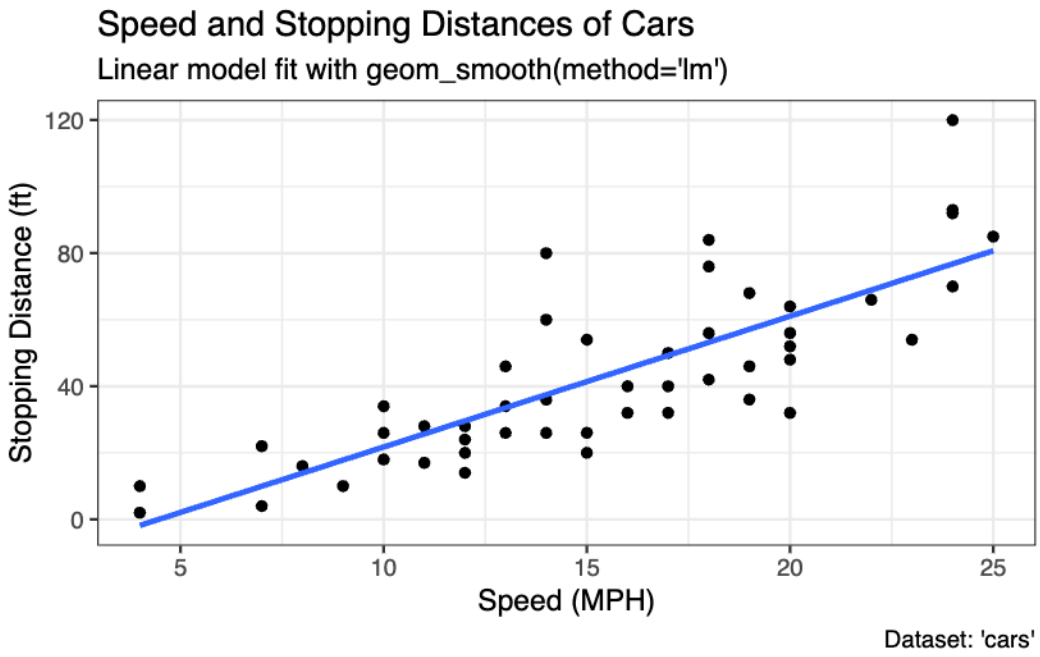
### Adding multiple geoms — geom_point() + geom_smooth()

We can layer geoms. Adding `geom_smooth()` on top of `geom_point()` shows the trend line.

```
# Two geoms: points + linear model trend line
ggplot(cars) +
  aes(x = speed, y = dist) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title    = "Speed and Stopping Distances of Cars",
    subtitle = "Linear model fit with geom_smooth(method='lm')",
    x        = "Speed (MPH)",
    y        = "Stopping Distance (ft)",
    caption  = "Dataset: 'cars'"
  ) +
  theme_bw()
```




`method = "lm"` fits a straight linear model line. `se = FALSE` removes the shaded confidence ribbon.

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### Section: Reading an Input File

We read in the DESeq2 differential expression results using `read.delim()`.

```
# Read gene expression data from course URL using read.delim()
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)

# Preview the data
head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
```

```
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

```
# How many genes are in this dataset?
nrow(genes)
```

```
[1] 5196
```

```
# How many columns?
ncol(genes)
```

```
[1] 4
```

```
colnames(genes)
```

```
[1] "Gene"       "Condition1" "Condition2" "State"
```

```
# How many up/down/unchanging genes?
table(genes$State)
```

```
      down unchanging         up
        72       4997        127
```

There are 5196 genes total. The `State` column classifies each gene as up-regulated, down-regulated, or unchanging.

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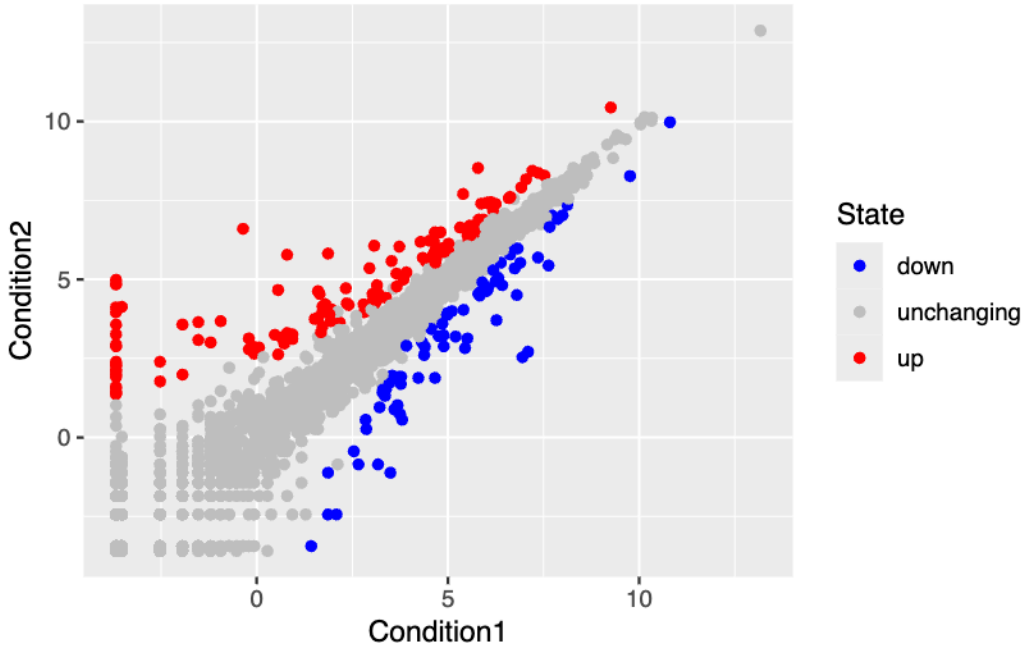
## Section: Plot with Custom Color Settings

Map the `State` column to color using `scale_colour_manual()` for explicit control over colors.

```
# Store base plot as object p
p <- ggplot(genes) +
  aes(x = Condition1, y = Condition2, col = State) +
  geom_point()

# Add custom colors with scale_colour_manual()
p + scale_colour_manual(values = c("blue", "gray", "red"))
```




Blue = down-regulated, gray = unchanging, red = up-regulated.

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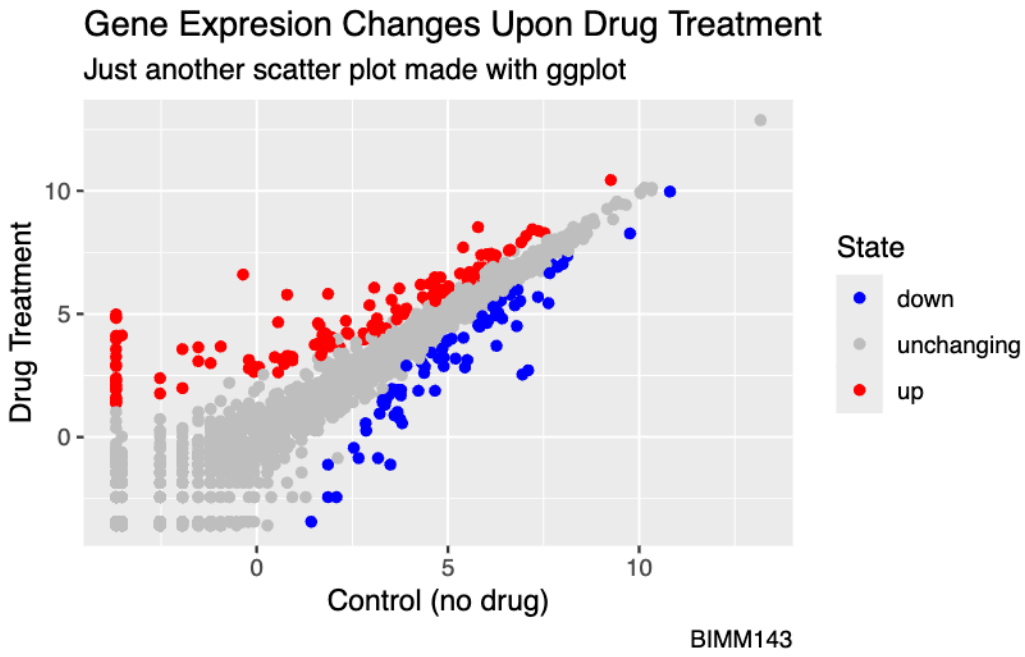
### Section: Plot with labs() Settings

Add descriptive labels to the plot using `labs()`.

```
# Final polished plot with custom colors and labs() annotations
p + scale_colour_manual(values = c("blue", "gray", "red")) +
  labs(
    title    = "Gene Expresion Changes Upon Drug Treatment",
    subtitle = "Just another scatter plot made with ggplot",
    x        = "Control (no drug)",
    y        = "Drug Treatment",
    caption  = "BIMM143"
  )
```

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## Section 7: Gapminder Dataset

The `gapminder` dataset contains economic and demographic data across countries since 1952.

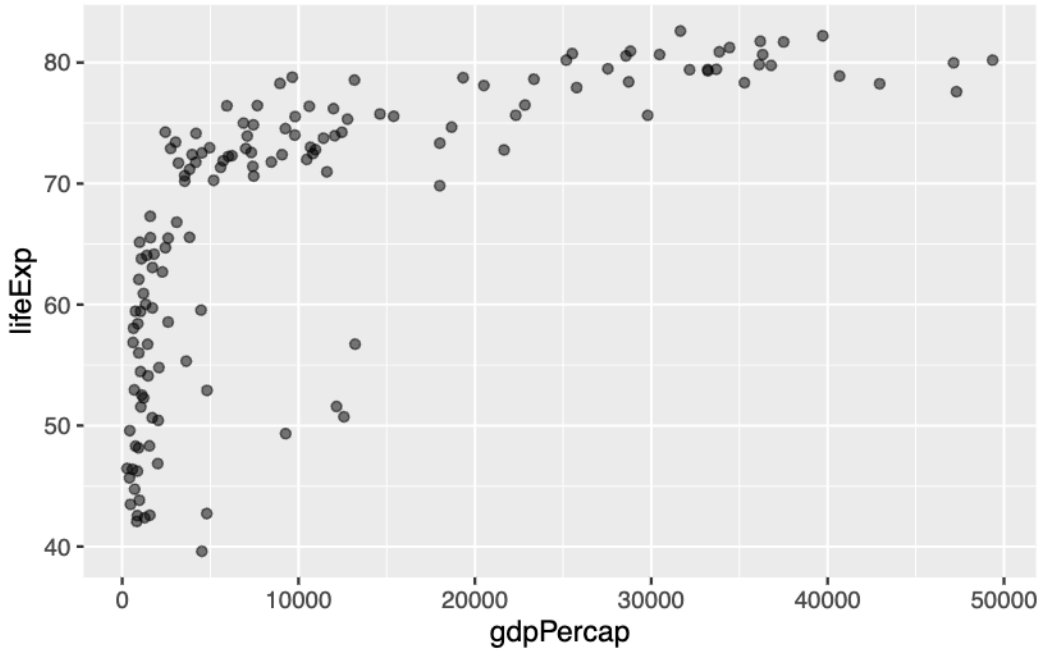
```
# install.packages("gapminder")
# install.packages("dplyr")
library(gapminder)
library(dplyr)

# Filter to year 2007
gapminder_2007 <- gapminder %>% filter(year == 2007)

head(gapminder_2007)
```

```
# A tibble: 6 x 6
  country     continent  year lifeExp      pop gdpPercap
  <fct>       <fct>     <int>   <dbl>    <int>     <dbl>
1 Afghanistan Asia       2007    43.8 31889923      975.
2 Albania     Europe     2007    76.4  3600523     5937.
```

```
3 Algeria     Africa     2007    72.3 33333216     6223.
4 Angola      Africa     2007    42.7 12420476     4797.
5 Argentina   Americas   2007    75.3 40301927    12779.
6 Australia   Oceania    2007    81.2 20434176    34435.
```

### Basic gapminder scatter plot

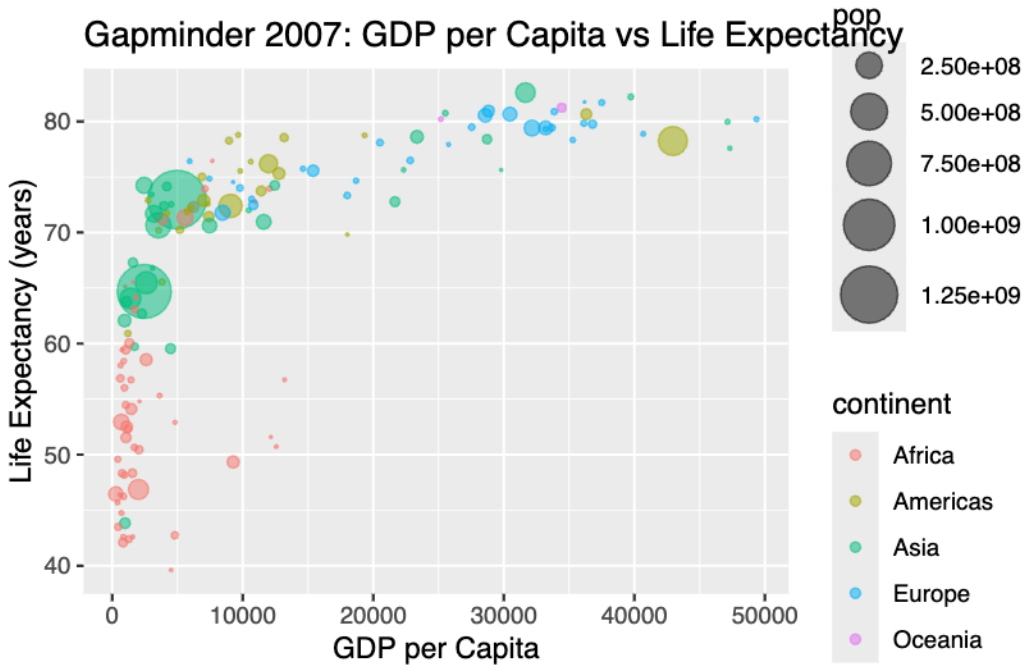
```r
# GDP per capita vs life expectancy for 2007
ggplot(gapminder_2007) +
  aes(x = gdpPercap, y = lifeExp) +
  geom_point(alpha = 0.5)
```




### Multi-variable plot with color and size

```r
# Map continent to color, population to point size
ggplot(gapminder_2007) +
  aes(x = gdpPercap, y = lifeExp, color = continent, size = pop) +
  geom_point(alpha = 0.5) +
  scale_size_area(max_size = 10) +
```

```
  labs(
    title = "Gapminder 2007: GDP per Capita vs Life Expectancy",
    x     = "GDP per Capita",
    y     = "Life Expectancy (years)"
  )
```




Each point is a country. Color = continent, size = population. Wealthier countries tend to have higher life expectancy.

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## Section 8 (Optional): Bar Charts

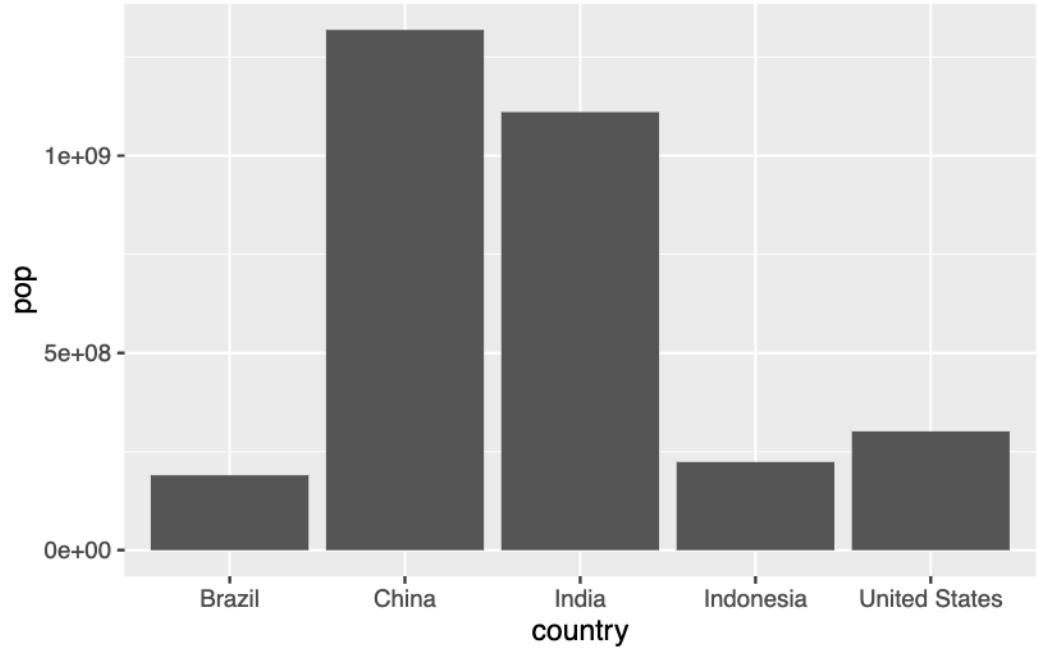
### Creating bar charts with geom_col()

```
# Get top 5 most populous countries in 2007
gapminder_top5 <- gapminder %>%
  filter(year == 2007) %>%
  arrange(desc(pop)) %>%
  top_n(5, pop)
```

```
gapminder_top5
```

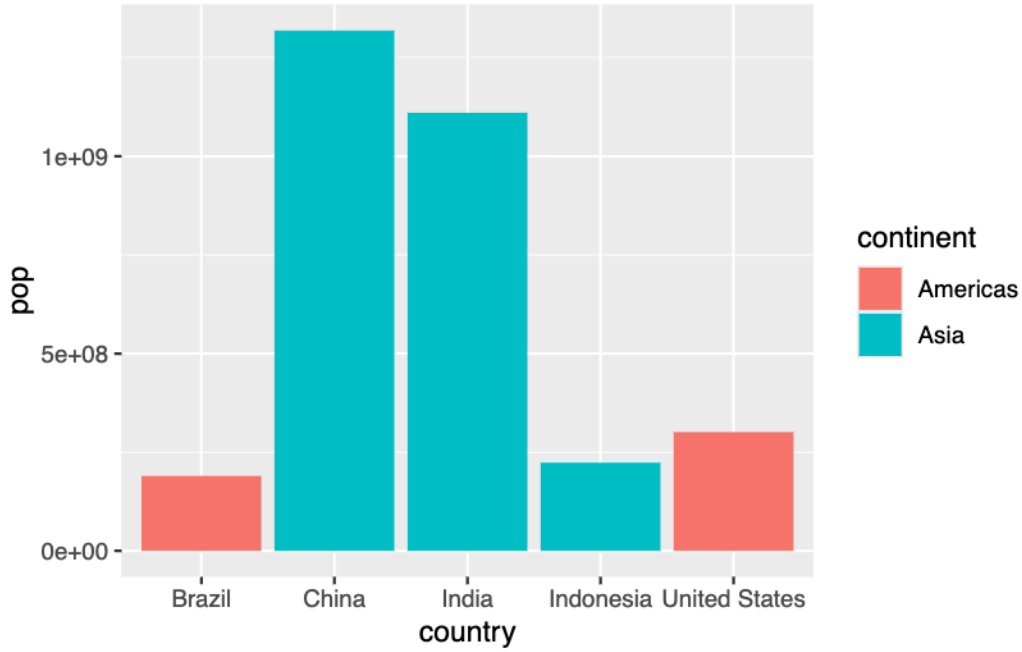
```
# A tibble: 5 x 6
  country       continent  year lifeExp        pop gdpPercap
  <fct>         <fct>     <int>   <dbl>      <int>     <dbl>
1 China         Asia       2007    73.0 1318683096     4959.
2 India         Asia       2007    64.7 1110396331     2452.
3 United States Americas   2007    78.2  301139947    42952.
4 Indonesia     Asia       2007    70.6  223547000     3541.
5 Brazil        Americas   2007    72.4  190010647     9066.
```

```
# Simple bar chart
ggplot(gapminder_top5) +
  geom_col(aes(x = country, y = pop))
```




**Filling bars with color**

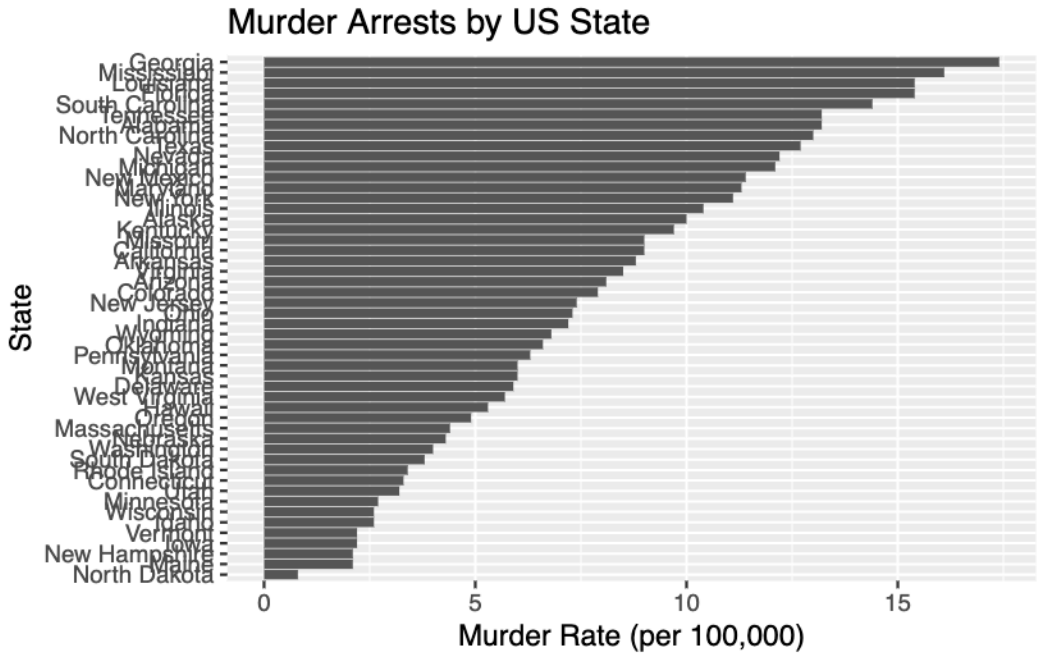
```
# Fill bars by continent (categorical variable)
ggplot(gapminder_top5) +
  geom_col(aes(x = country, y = pop, fill = continent))
```




**Flipping bar charts with coord_flip()**

```
# US Arrests dataset - reorder states by murder rate and flip
USArrests$State <- rownames(USArrests)

ggplot(USArrests) +
  aes(x = reorder(State, Murder), y = Murder) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Murder Arrests by US State",
    x     = "State",
    y     = "Murder Rate (per 100,000)"
  )
```




`coord_flip()` rotates the chart horizontal, making state labels readable.

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## Session Info

```
sessionInfo()
```

```
R version 4.5.3 (2026-03-11)
Platform: aarch64-apple-darwin20
Running under: macOS Tahoe 26.4.1

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/Los_Angeles
tzcode source: internal
```

```
attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
[1] dplyr_1.2.1     gapminder_1.0.1 ggplot2_4.0.2

loaded via a namespace (and not attached):
 [1] vctrs_0.7.2        nlme_3.1-168       cli_3.6.5          knitr_1.51
 [5] rlang_1.1.7        xfun_0.57          generics_0.1.4     S7_0.2.1
 [9] jsonlite_2.0.0     labeling_0.4.3     glue_1.8.0         htmltools_0.5.9
[13] scales_1.4.0       rmarkdown_2.31     grid_4.5.3         evaluate_1.0.5
[17] tibble_3.3.1       fastmap_1.2.0      yaml_2.3.12        lifecycle_1.0.5
[21] compiler_4.5.3     RColorBrewer_1.1-3 pkgconfig_2.0.3    mgcv_1.9-4
[25] lattice_0.22-9     farver_2.1.2       digest_0.6.39      R6_2.6.1
[29] utf8_1.2.6         tidyselect_1.2.1   splines_4.5.3      pillar_1.11.1
[33] magrittr_2.0.5     Matrix_1.7-4       withr_3.0.2        tools_4.5.3
[37] gtable_0.3.6
```